EECS 498/CSE 598: Zero-Knowledge Proofs
Winter 2026
Lecture 12

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Agenda for this lecture

- Announcements
- Random oracles are collision-resistant
- Security of Fiat-Shamir
- “Weak” Fiat-Shamir attacks
- Zero knowledge: intuition

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Announcements

- Project 2 is online
- Autograder is working
- Due date: 2/20
  - <u>May</u> get pushed back – will announce if so

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Collision-resistance of RO

$\underline{\mathsf{CollRO}(\mathcal{A})}$:

$T \leftarrow [\,]$

$x_1, x_2 \leftarrow\$ \mathcal{A}^{\mathbf{F}}$

Return $\mathbf{F}(x_1) = \mathbf{F}(x_2) \wedge x_1 \neq x_2$

$\underline{\mathbf{F}(x)}$:

If $T[x] = \perp$:

  $y \leftarrow\$ \{0,1\}^n$

  $T[x] \leftarrow y$

Return $T[x]$

Collision-resistance of RO

$\mathsf{CollRO}(\mathcal{A})$:

$T \leftarrow [\,]$

$x_1, x_2 \leftarrow\$ \mathcal{A}^{\mathbf{F}}$

Return $\mathbf{F}(x_1) = \mathbf{F}(x_2) \wedge x_1 \neq x_2$

$\mathbf{F}(x)$:

If $T[x] = \perp$:

$\quad y \leftarrow\$ \{0,1\}^n$

$\quad T[x] \leftarrow y$

Return $T[x]$

Theorem. *For any adversary $\mathcal{A}$ making at most q queries to $\mathbf{F}$ in game* CollRO,

$$\Pr[\mathsf{CollRO}(\mathcal{A}) = 1] \leq \frac{q^2}{2^n}.$$

Proof:

Create new game G

G is CollRO except

$F(X)$:

If $T[X] = \perp$

$\quad y \leftarrow\$ \{0,1\}^n$

If $\exists X_0$ with y as eval

$\quad y \leftarrow\$ \{0,1\}^n \setminus T$

$T[X] \leftarrow y$

Ret $T[X]$

$\Pr[G(\mathcal{A}) = 1] = 0$

$\Pr[\mathsf{CollRO}(\mathcal{A}) = 1]$

$\leq \Pr[G(\mathcal{A}) = 1] + \Pr[\text{resample}]$

$\Pr[\text{resample}] \leq \frac{q^2}{2^n}$ $\boxtimes$

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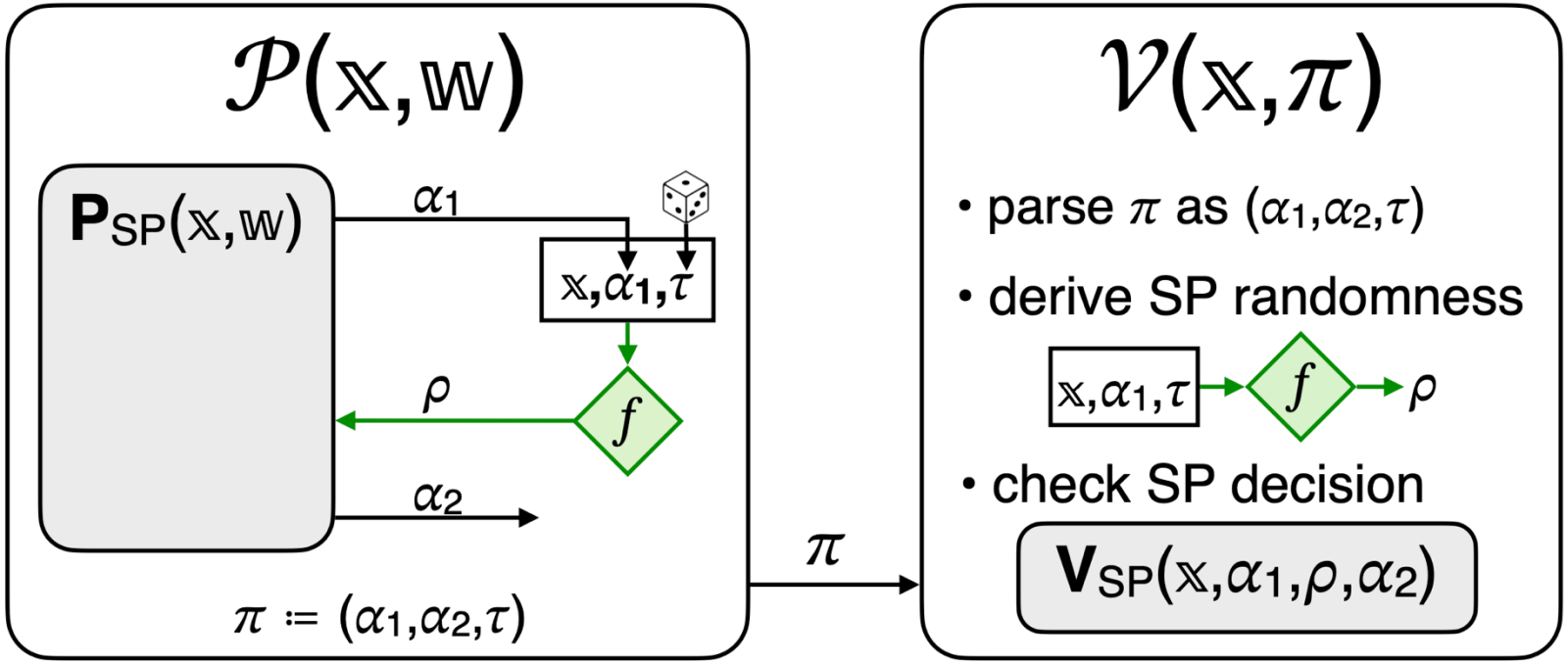
Fiat-Shamir: replace verifier with RO



(Courtesy Chiesa-Yogev)

Security for F-S

Thm:

Let $Arg = \langle Setup, \mathbb{P}, \mathbb{V} \rangle$ be an interactive argument for relation R. Let $FS^f[Arg]$ be the F-S compiled non-interactive argument.

If Arg has state-restoration knowledge soundness with κ_{sr}, $FS^f[Arg]$ has adaptive (NI) knowledge soundness $\kappa_{FS} \leq \kappa_{sr}$

Proof:

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”Weak” Fiat-Shamir attacks

$\mathsf{NISchnorr}.\mathcal{P}^{\mathbf{F}}(\mathbb{G}, G, p, X; x)$:

$r \leftarrow\$ \mathbb{Z}_p \ ; \ R \leftarrow rG$

$c \leftarrow \mathbf{F}(\mathbb{G}, G, p, X, R)$

$z \leftarrow r + cx$

Return (R, z)

$\mathsf{NISchnorr}.\mathcal{V}^{\mathbf{F}}(\mathbb{G}, G, p, X, \pi)$:

$R, z \leftarrow \pi$

$c \leftarrow \mathbf{F}(\mathbb{G}, G, p, X, R)$

Return $zG = R + cX$

”Weak” Fiat-Shamir attacks

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$\mathsf{WFSSchnorr}.\mathcal{P}^{\mathbf{F}}(\mathbb{G}, G, p, X; x)$:

$r \leftarrow\$ \mathbb{Z}_p \; ; \; R \leftarrow rG$

$c \leftarrow \mathbf{F}(R)$

$z \leftarrow r + cx$

Return $X, (R, z)$

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”Weak” Fiat-Shamir attacks

$\mathsf{WFSSchnorr}.\mathcal{P}^{\mathbf{F}}(\mathbb{G}, G, p, X; x)$:

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$z \leftarrow r + cx$

Return $X, (R, z)$

$\mathsf{WFSSchnorr}.\mathcal{V}^{\mathbf{F}}(\mathbb{G}, G, p, X, \pi)$:

$R, z \leftarrow \pi$

$\boxed{c \leftarrow \mathbf{F}(R)}$

Return $zG = R + cX$

Done this way, non-interactive Schnorr lacks *adaptive* knowledge soundness…

$A^F(\mathbb{G}, G, p)$:

$R \leftarrow\$ \mathbb{G}$; $c \leftarrow F(R)$

$z \leftarrow\$ \mathbb{Z}_p$

Ret $(zc^{-1}G - c^{-1}R), (R, z)$

$zG = R + cX$

$zG = R + c(zc^{-1}G - c^{-1}R)$

$F(R)$

Delphian has weak F-S

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Zero knowledge: starting with intuition

PIOP $\xrightarrow{PC}$ Argument $\xrightarrow{F-S}$ NI-ArgoK

SNARK

Get to ZKSNARK via

third protocol transform

ZK means witness is hidden

In your own words?

V can't learn $f(w)$ for any f

Special HVZK for sigma protocols

Definition. *A sigma protocol* $\mathsf{IP} = \langle \mathcal{P}, \mathcal{V} \rangle$ *(with verification function* verify*) for the language* $\mathcal{L}$ *is special honest-verifier zero-knowledge if there exists a probabilistic polynomial time simulator* $\mathcal{S}$ *that takes parameters* pp*, input* x *and challenge* $c \in \mathcal{C}$ *and satisfies:*

1. *For* $\mathsf{pp} \leftarrow_{\$} \mathsf{Setup}$ *and all pairs* x, c*, the output of* $\mathcal{S}(\mathsf{pp}, x, c)$ *is a proof* π *such that*

$$\mathsf{verify}(\mathsf{pp}, x, R, c, z) = 1$$

.

2. *For all* $(x, w) \in \mathcal{R}_{\mathcal{L}}$*, the transcript* R, c, z *computed as*

$$c \leftarrow_{\$} \mathcal{C} \ ; \ R, z \leftarrow_{\$} \mathcal{S}(x, c)$$

is identically distributed to a real transcript of the interaction $[\mathcal{P}(\mathsf{pp}, x, w) \leftrightarrow \mathcal{V}(\mathsf{pp}, x)]$.

Special HVZK for Schnorr

Theorem: The Schnorr protocol is special HVZK.